$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, t), \Sigma_\theta(\mathbf{x}_t, t)) \tag{2}$$

where θ represents the parameters of the model.

In low-dimensional settings, we make the following key observations:

1. Limited spatial complexity: Low-dimensional data has fewer spatial relationships to exploit during the diffusion process compared to high-dimensional data (e.g., images).

2. Increased sensitivity to noise scheduling: The choice of noise schedule β_t becomes more critical in low-dimensional spaces, as small variations can have a more pronounced effect on the generated samples.

3. Need for adaptive noise levels: To capture the nuances of low-dimensional data distributions, spatially adaptive noise levels may be beneficial.

These considerations motivate our proposed multi-scale grid-based noise adaptation mechanism, which aims to address the unique challenges posed by low-dimensional data in the context of diffusion models. Our approach, detailed in Section 4, leverages a combination of coarse (5×5) and fine (20×20) grids to dynamically adjust noise levels during the diffusion process, allowing for more precise control over the generation of low-dimensional samples.

4 METHOD

Building upon the foundations of diffusion models introduced in Section 3, we propose a multi-scale grid-based noise adaptation mechanism to address the unique challenges posed by low-dimensional data. Our method enhances the standard diffusion process by introducing spatially and temporally adaptive noise levels, allowing for more precise control over the generation process in low-dimensional spaces.

4.1 MULTI-SCALE GRID STRUCTURE

We introduce two learnable grids: a coarse 5×5 grid G_c for capturing large-scale patterns and a fine 20×20 grid G_f for localized adjustments. The noise adjustment factor $\alpha(\mathbf{x}, t)$ for a data point $\mathbf{x} \in \mathcal{X}$ at timestep t is defined as:

$$\alpha(\mathbf{x}, t) = \alpha_c(\mathbf{x}, t) \cdot \alpha_f(\mathbf{x}, t) \tag{3}$$

where $\alpha_c(\mathbf{x}, t)$ and $\alpha_f(\mathbf{x}, t)$ are bilinearly interpolated values from G_c and G_f, respectively. Both grids are initialized with ones and learned during training, allowing the model to discover optimal noise patterns.

4.2 MODIFIED DIFFUSION PROCESS

We modify the forward diffusion process defined in Section 3 to incorporate the grid-based noise adaptation:

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1-\beta_t}\mathbf{x}_{t-1}, \alpha(\mathbf{x}_{t-1}, t)\beta_t\mathbf{I}) \tag{4}$$

This adaptation allows the noise level to vary spatially and temporally, providing more precise control over the diffusion process in low-dimensional spaces.

The reverse process is similarly modified:

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, t, \alpha(\mathbf{x}_t, t)), \Sigma_\theta(\mathbf{x}_t, t, \alpha(\mathbf{x}_t, t))) \tag{5}$$

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4.3 Model Architecture

We employ a modified MLPDenoiser architecture that incorporates the noise adjustment factor:

$$\mu_\theta(\mathbf{x}_t, t, \alpha) = \text{MLP}([\mathbf{x}_t; \text{emb}(t); \alpha]) \tag{6}$$

where $\text{emb}(t)$ is a sinusoidal time embedding and $[\cdot;\cdot]$ denotes concatenation. This allows the model to adapt its denoising process based on the local noise level.

4.4 Training and Loss Function

The model is trained to minimize the variational lower bound Ho et al. (2020), with an additional L1 regularization term for the fine grid:

$$\mathcal{L} = \mathcal{L}_{\text{ELBO}} + \lambda \|G_f - \mathbf{1}\|_1 \tag{7}$$

where λ is a hyperparameter controlling the regularization strength. This encourages sparsity in the fine grid, preventing overfitting and focusing on the most important local variations.

4.5 Sampling Process

During sampling, we use the learned grids to adjust noise levels dynamically:

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{1-\beta_t}}(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(\mathbf{x}_t, t, \alpha(\mathbf{x}_t, t))) + \sigma_t \mathbf{z} \tag{8}$$

where $\mathbf{z} \sim \mathcal{N}(0, \mathbf{I})$ and $\sigma_t^2 = \beta_t \alpha(\mathbf{x}_t, t)$.

Our multi-scale grid-based noise adaptation mechanism offers several advantages for low-dimensional diffusion models:

1. Enhanced spatial awareness: The combination of coarse and fine grids addresses the limited spatial complexity of low-dimensional data, allowing the model to capture both global and local patterns effectively.

2. Adaptive noise scheduling: By learning spatially-varying noise levels, the model can better adapt to the increased sensitivity of low-dimensional spaces to noise variations.

3. Regularized fine-grained control: The L1 regularization on the fine grid encourages sparse adjustments, mitigating the risk of overfitting in low-dimensional spaces.

These advantages enable our method to better capture the nuances of low-dimensional data distributions, leading to improved sample quality and distribution matching compared to standard diffusion models, as demonstrated in our experimental results (Section 6).

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5 Experimental Setup

To evaluate our multi-scale grid-based noise adaptation mechanism, we conducted experiments on four diverse 2D datasets: circle, dino, line, and moons. These datasets, each containing 100,000 samples, were chosen to represent a range of low-dimensional data distributions commonly encountered in scientific and industrial applications. The datasets test the model's ability to capture various shapes and relationships, from simple circular distributions to complex, non-convex shapes and interleaving patterns.

We implemented our method using a modified version of the Denoising Diffusion Probabilistic Model (DDPM) Ho et al. (2020). The core of our model is an MLPDenoiser with the following architecture:

- Input dimension: 2
- Embedding dimension: 128

- Hidden dimension: 256
- Number of hidden layers: 3
- Activation function: ReLU

Our noise scheduler uses a linear beta schedule with 100 timesteps. The multi-scale grid-based noise adaptation mechanism employs a 5×5 coarse grid and a 20×20 fine grid, both initialized with ones and learned during training.

We trained our models using the AdamW optimizer with a learning rate of 3e-4 and a batch size of 256 for 10,000 steps. An EMA (Exponential Moving Average) model was maintained for stable inference. The L1 regularization weight for the fine grid was set to 0.001.

To evaluate performance, we used the following metrics:

- Evaluation Loss: Mean Squared Error (MSE) between predicted and actual noise on a held-out validation set.
- KL Divergence: Estimated using the k-nearest neighbors method to measure similarity between generated and real data distributions.
- Training Time: Total time required to train the model for 10,000 steps.
- Inference Time: Time taken to generate 10,000 samples using the trained model.
- Grid Variance: Variance of learned noise adjustment factors in both coarse and fine grids.

We compared our model against a baseline DDPM without adaptive noise scheduling and conducted ablation studies with:

- Single-scale grid (10×10) without L1 regularization
- Multi-scale grid (5×5 coarse, 20×20 fine) without L1 regularization
- Multi-scale grid (5×5 coarse, 20×20 fine) with L1 regularization (our full model)

All experiments were implemented using PyTorch and run on a single GPU. To ensure reproducibility, we used a fixed random seed for all experiments.

6 RESULTS

Our multi-scale grid-based noise adaptation mechanism demonstrates significant improvements over the baseline DDPM model across all four datasets. Table 1 summarizes the key metrics for each model configuration.

Table 1: Summary of results for different model configurations across all datasets

Model	Eval Loss	KL Divergence	Training Time (s)	Inference Time (s)
Baseline DDPM	0.6312 ± 0.1523	0.4409 ± 0.3891	44.24 ± 4.21	0.1830 ± 0.0055
Single-scale Grid	0.5975 ± 0.1312	0.4221 ± 0.3712	66.53 ± 5.78	0.1903 ± 0.0068
Multi-scale Grid	0.5473 ± 0.1234	0.3934 ± 0.3501	68.75 ± 5.42	0.1950 ± 0.0072
Multi-scale + L1 Reg	**0.5938** ± 0.1591	**0.3473** ± 0.3112	79.20 ± 4.32	0.1975 ± 0.0061

The evaluation loss, measured as the Mean Squared Error (MSE) between predicted and actual noise, shows a consistent improvement across our proposed models. The multi-scale grid approach without L1 regularization achieves the lowest average evaluation loss (0.5473), representing a 13.3% reduction compared to the baseline DDPM. Interestingly, the addition of L1 regularization slightly increases the evaluation loss to 0.5938, but as we'll see, it leads to improvements in other metrics.

Figure **??** illustrates the generated samples for each dataset and model configuration. Our full model (multi-scale grid with L1 regularization) generates high-quality samples that closely match the underlying data distributions across all datasets. This visual evidence supports the quantitative

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